

# **HUMAN-AI COLLABORATION IN RADIOLOGY**

## The Case of Pulmonary Embolism

---

Paul Goldsmith-Pinkham (Yale SOM)

Chenhao Tan (Chicago CS/DS)

Alexander K. Zentefis (Santa Clara Leavey)

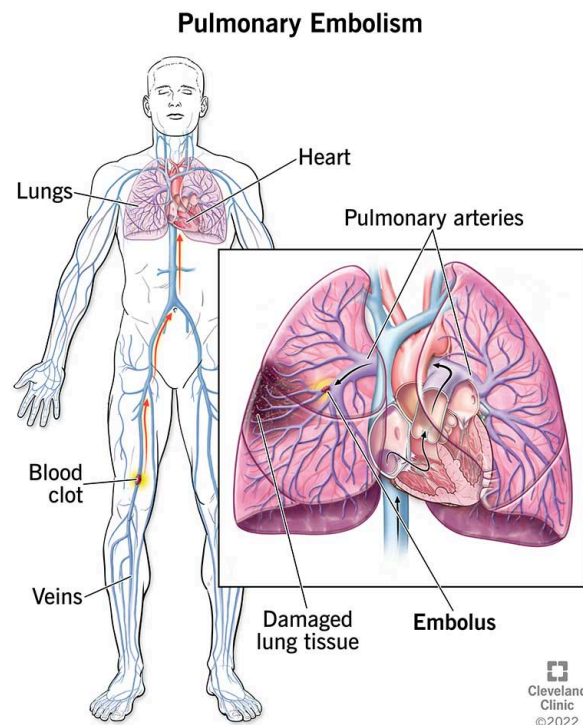
Yale IRB-2000037424    <https://paulgp.com>

- HR managers screen candidates with AI (Dattner et al. 2019)
- Financial analysts evaluate investments with AI (Roy et al. 2025)
- Judges assess recidivism risk with AI (Angelova et al. 2025)

Lots of evidence that the algorithms are **accurate**. Much less on what experts **do** with them.

**What does human-AI collaboration actually look like, day to day, in real practice?**

- PE = blood clot in pulmonary arteries
- **3rd cardiovascular cause of death** in U.S.; 60–100k deaths/yr
- Untreated mortality > 30%; treated < 10%
- Standard of care: CT pulmonary angiography (CTPA)
- Deep-learning on CTPA: sensitivity 0.88, specificity 0.86 (Soffer 2021)
- FDA-cleared decision-support tools now common



Track every CTPA at a large academic health system as an FDA-cleared AI tool rolls out.

## Setting

- 1 academic health system, 8 care sites
- Staggered rollout: Aug 2019 – Jul 2022
- 117,063 CTPAs, 84,640 patients
- 389 signing radiologists

## Linked at the scan level

- AI prediction (binary PE flag)
- Radiologist diagnosis from clinical reports
- Engagement (hovers, timestamps)
- Patient outcomes (mortality, complications)
- NPI registry → rad demographics

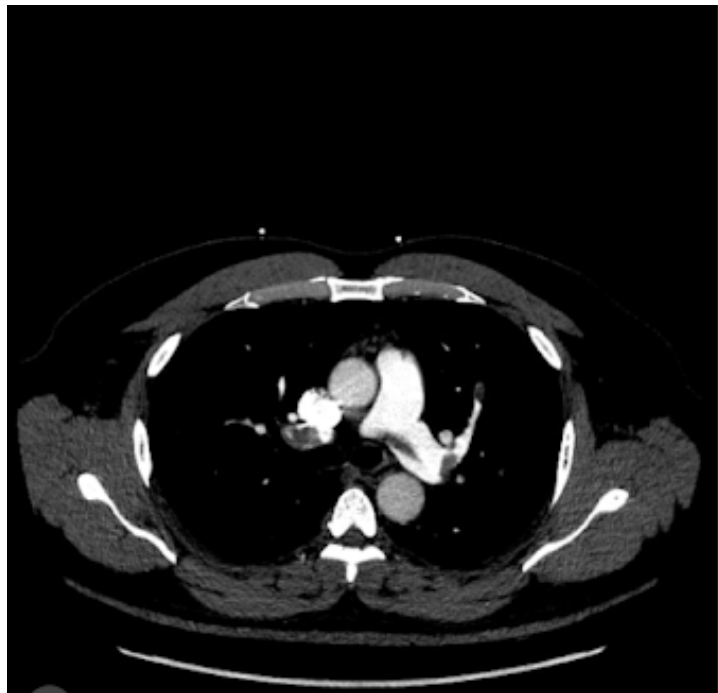
Complementary to lab-in-the-field experiments (*Agarwal, Moehring, Rajpurkar, Salz 2023*) — actual clinical practice over five years.

1. **Asymmetric agreement.** Radiologists agree with AI on 97% of negatives, only 84% of positives
2. **Convergence.** Override rate on AI+ falls 30% → 12% in two years, then stabilizes
3. **Stable speed despite higher load.** Time per scan flat at 3.6 hr; per-radiologist volume nearly doubles
4. **Persistent heterogeneity.** Female rads override 6 pp less; engagement is non-monotonic
5. **Asymmetric resolution on follow-ups.** 54% of rad-overrides resolve to “no PE”; 0% of rad-rejections of AI+ persist
6. **Outcomes differ by agreement category** — but mortality differences are modest

1. Setting and data
2. Patterns of (dis)agreement
3. What follow-up scans tell us
4. Patient outcomes by agreement category
5. Conclusion + ongoing causal work

# **1. SETTING AND DATA**

---



**Direct sign:** filling defect in pulmonary artery

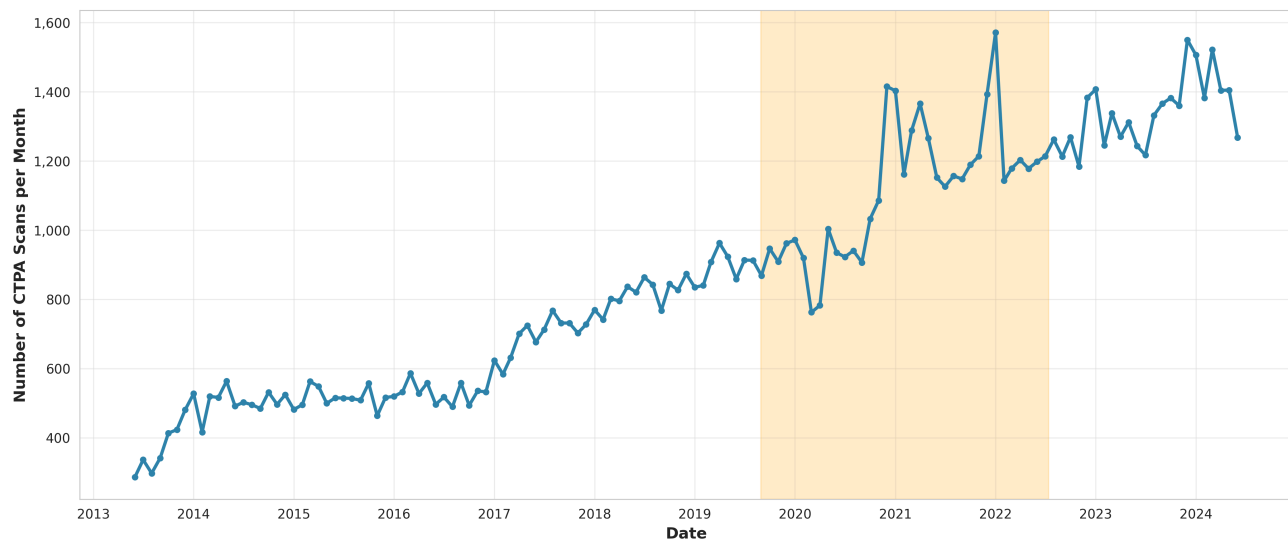


**Indirect sign:** RV/LV = 1.37 → right heart strain

Hundreds of cross-sectional slices per scan. AI flags candidate slices.



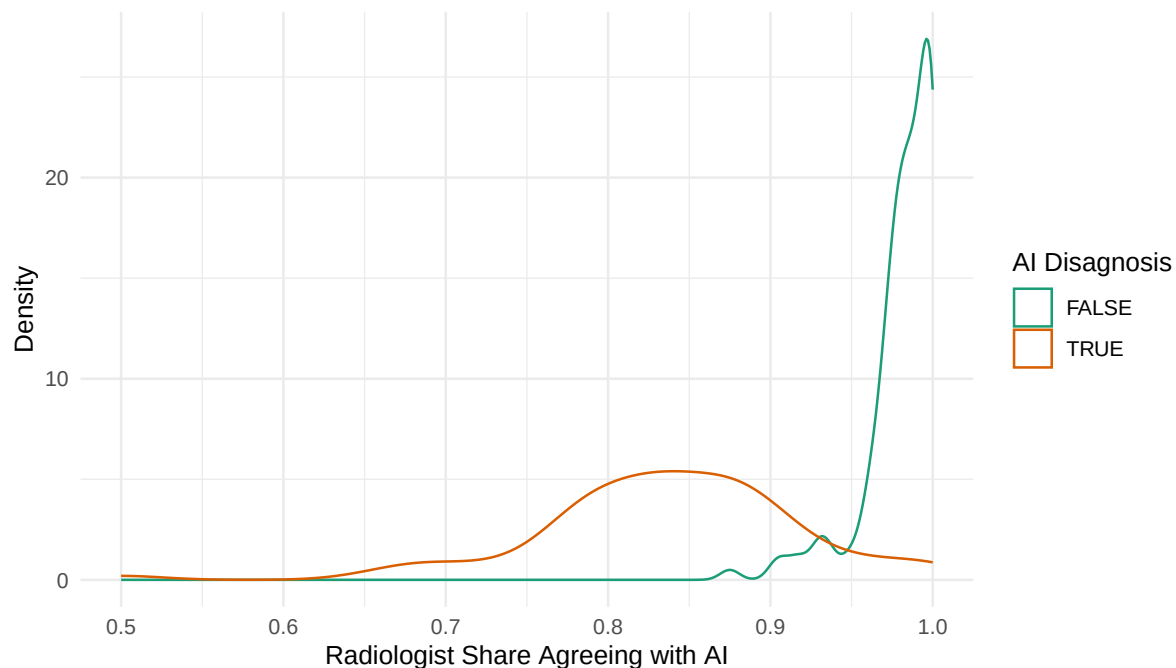
- FDA-cleared CNN, trained on 15,000+ annotated CTPAs
- Binary output per scan: **PE detected** / **no PE**
- When AI flags PE:
  - Scan moves to the **top** of the worklist
  - Bright orange alert + heatmap on the suspect slice
- When AI does not flag PE:
  - Grey indicator, scan keeps queue position
- Core functionality stable across the sample; UI improves over time  
[\[Engagement metrics\]](#)



- Orange band: AI rollout window
- Volume +16% during/after rollout
- 8 sites, rolled in Aug 2019 – Jul 2022
- PE positivity stable at ~ 10% throughout  
[Positivity]
- [Care sites] · [Patients] · [Scans]

## **2. PATTERNS OF (DIS)AGREEMENT**

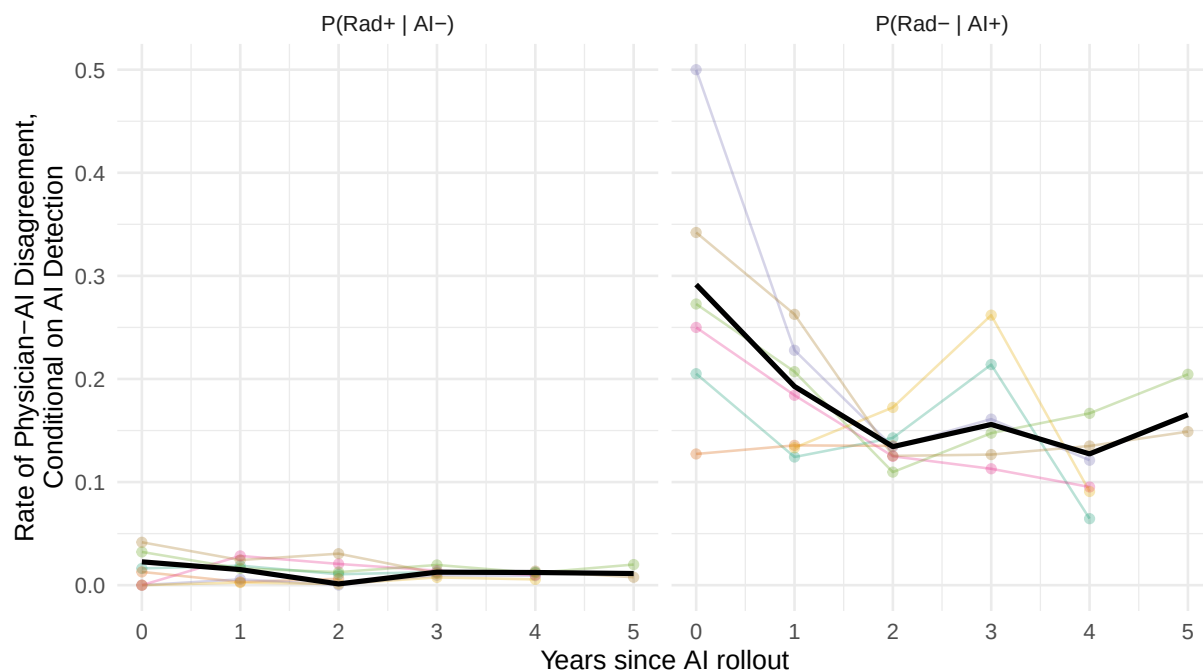
---



- $P(\text{Rad} + \mid \text{AI} +) = 84.1\%$
- $P(\text{Rad} - \mid \text{AI} -) = 97.2\%$
- 13 pp gap between the two
- Rads override AI+ 16% of the time, override AI- only 3% of the time

**Asymmetric trust** in algorithmic recommendations

# Override on AI+ falls 30% → 12% in two years, then stabilizes



- **Right panel:** override on AI+ falls 30% → 12% by year 2, then flat
- **Left panel:** override on AI- stable at 2–3%

Consistent with:

- **Learning,** or
- **Automation bias**

[ED only]

## Female – Male radiologists

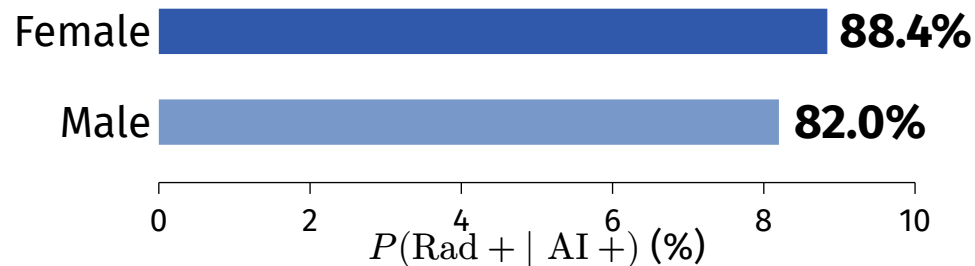
- $P(\text{Rad} + | \text{AI} +)$ : 88.4% vs 82.0%
- Difference: **+6.4 pp** ( $p = 0.037$ )
- Female rads override AI+ less often

## Experience (years since med school)

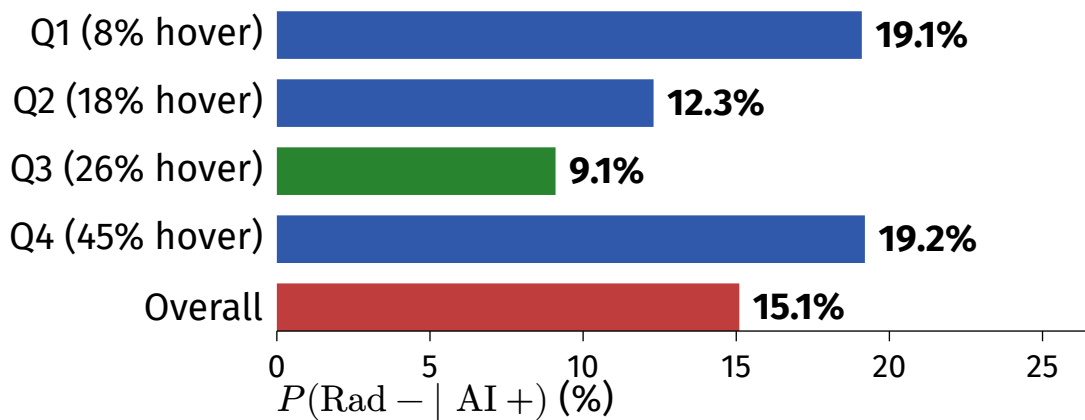
- $F = 0.79$ ,  $p = 0.50$
- No precise linear trend ( $p = 0.24$ )

## Scan volume quartiles

- $F = 0.09$ ,  $p = 0.97$
- No precise differences



[Full demographics] [Shift / weekend]



## Two modes of disagreement

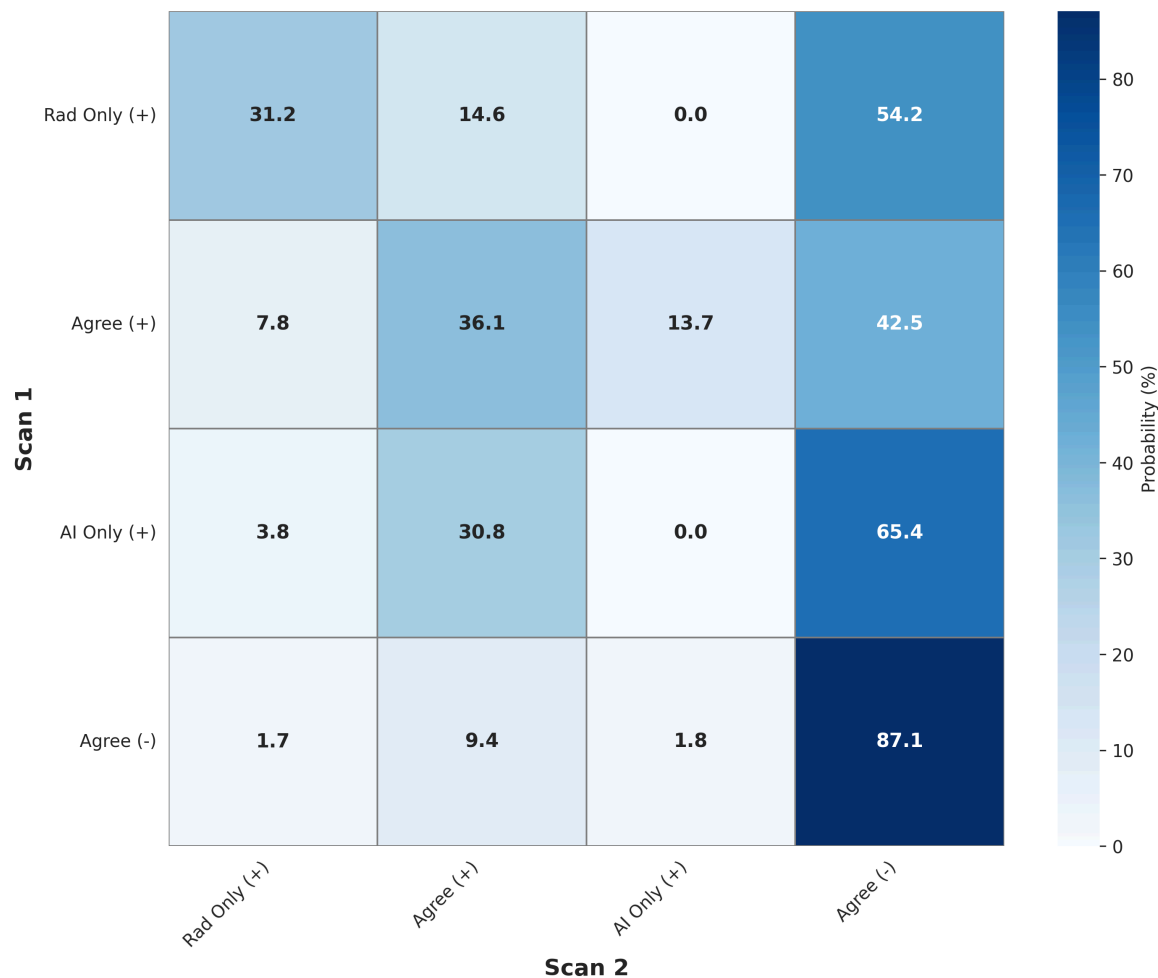
- **Q1: passive bypass.** Low hover, low engagement → rely on own judgment, skip AI
- **Q3: sweet spot.** Moderate hover ( 26%) → highest agreement
- **Q4: active vetting.** High hover ( 45%) → look at AI evidence, find reasons to reject

Disagreement at low and high engagement **is not the same behavior.**

### **3. WHAT FOLLOW-UP SCANS TELL US**

---

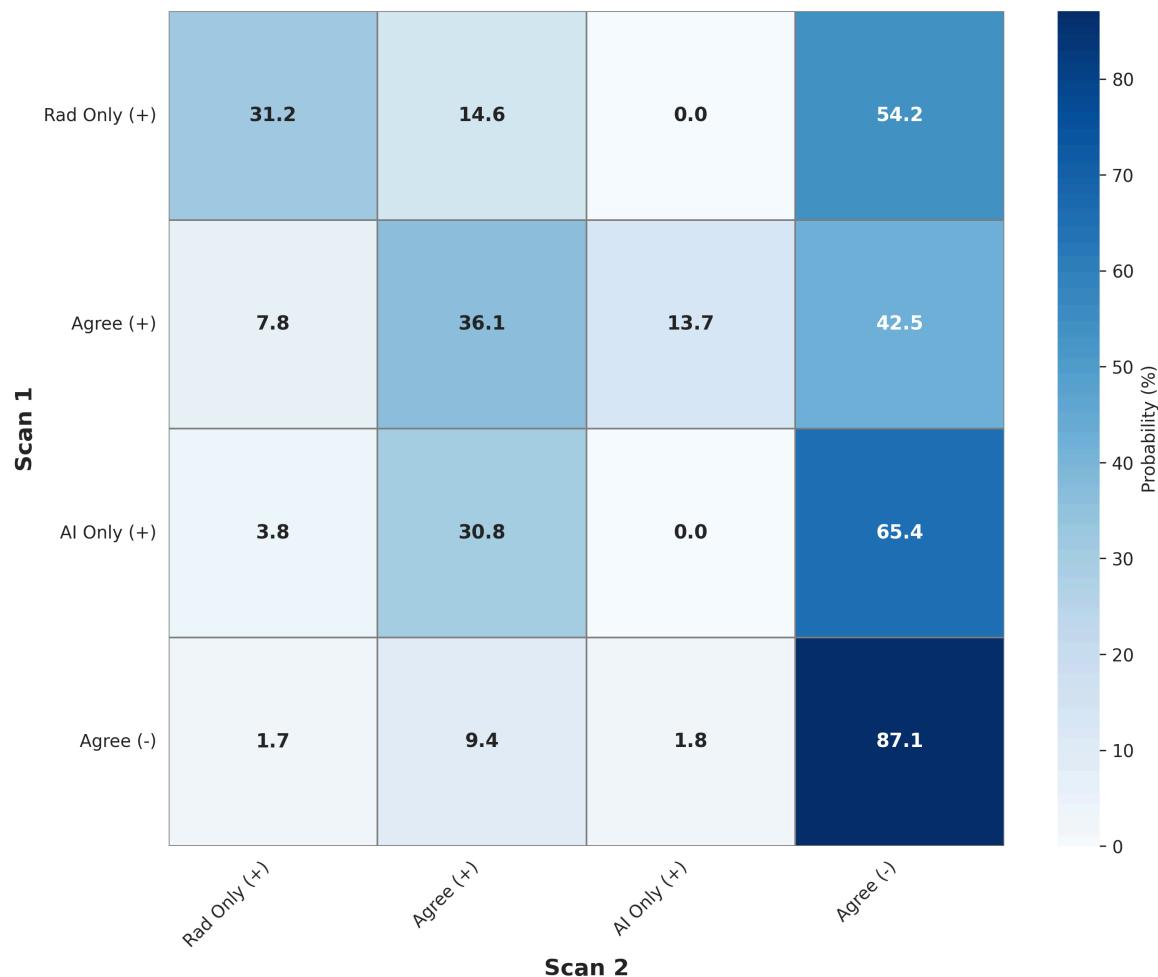




Among 35,896 first scans, 1,375 (3.8%) return within 30 days.

**Rad Only (+)** — rad overrides AI to call PE:

- 31% persist as Rad Only (+)
- **54% resolve to Agree (-)**



Among 35,896 first scans, 1,375 (3.8%) return within 30 days.

**AI Only (+)** — rad rejects AI flag:

- **0% persist**
- 65% → Agree (-); 31% → Agree (+)

Override-to-diagnose-PE ≠ override-to-reject-PE.

[Unconditional heatmap]

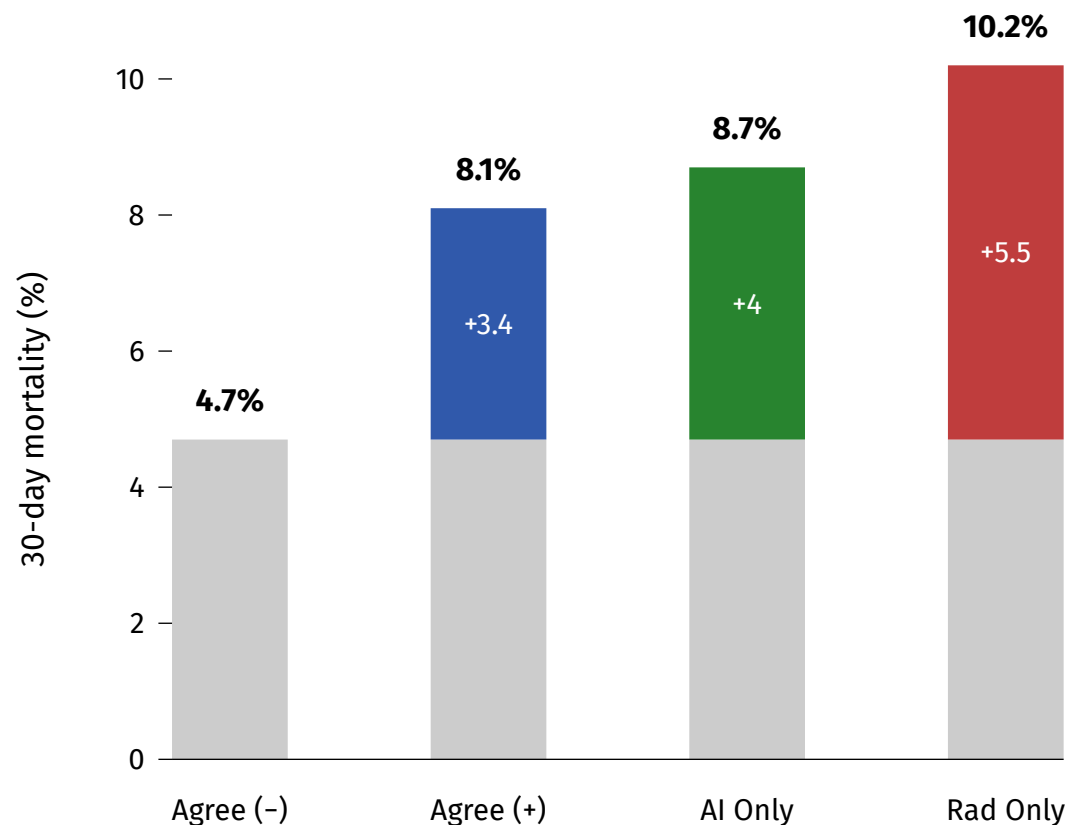
## **4. PATIENT OUTCOMES BY AGREEMENT CATEGORY**

---

$$Y_i = \beta_0 + \beta_1 \underbrace{\text{Agree}(+)_i}_{\text{both detect}} + \beta_2 \underbrace{\text{AI Only}(+)_i}_{\text{AI flags, rad rejects}} + \beta_3 \underbrace{\text{Rad Only}(+)_i}_{\text{rad overrides AI}} + \gamma_s + \delta_{hd} + \lambda_t + \varepsilon_i$$

- Reference category: **Agree (-)** — both find no PE
- $\gamma_s$ : care site FE;  $\delta_{hd}$ : hour-of-day  $\times$  day-of-week FE;  $\lambda_t$ : year FE
- First scans only ( $N = 35,851$ ); heteroskedasticity-robust SEs

**Read  $\beta_k$  as:** % of patients with outcome, relative to “both agree no PE”

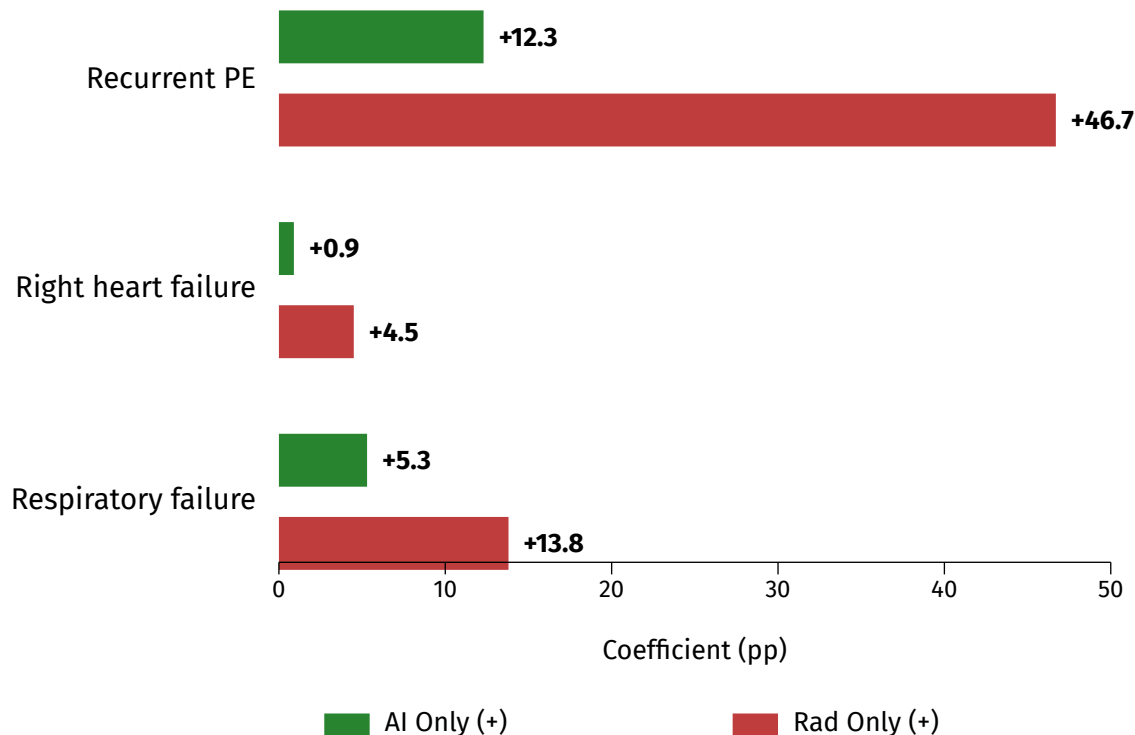


## 30-day mortality

- Baseline: 4.7%
- Agree (+): +3.4 pp
- AI Only (+): +4.0 pp
- Rad Only (+): +5.5 pp

Same ordering at **90 days**  
(+4.2 / +5.9 / +7.8) and **1 year**  
(+4.8 / +7.1 / +9.2).

AI Only vs Rad Only: not  
precisely estimated for  
mortality.



- **Recurrent PE:** +47 pp for Rad Only vs +12 pp for AI Only ( $t = -12.7$ )
- **Right heart failure:** +4.5 vs +0.9 ( $t = -3.3$ )
- **Respiratory failure:** +13.8 vs +5.3 ( $t = -3.0$ )

Rad-overrides **behave like real PE** in downstream care: anticoagulation, follow-up imaging, complications. AI+ rejected by rads do not.

1. **Asymmetric agreement.** Rads vet AI positives (84%) but accept AI negatives (97%)
2. **Sharp convergence, persistent heterogeneity.** Override on AI+ falls 30% → 12%, then flat — but right tail rebounds
3. **Productivity through workflow, not rushed reads.** Per-scan time stable; per-rad volume nearly doubles
4. **Engagement is non-monotonic.** Low and high hover rates both look like more disagreement — for different reasons
5. **Disagreement direction matters downstream.** Rad-overrides resolve more often, but track real PE in care; AI+ rejected by rads do not

Collaboration is **active judgment**, not passive acceptance or rejection.

Staggered rollout × quasi-random patient–radiologist assignment lets us identify:

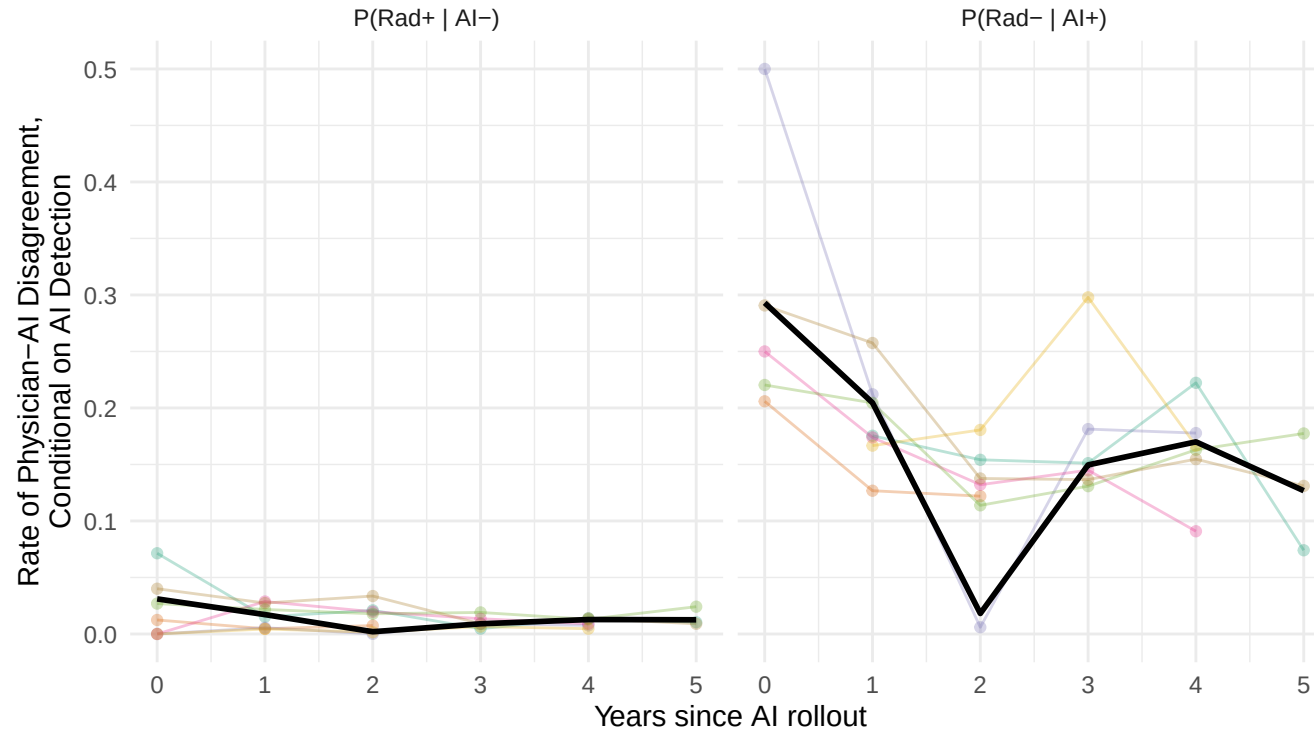
1. Effect of AI on radiologist **diagnostic skill** at suspected PE
2. Effect on **false-positive vs. false-negative trade-off** (preferences)
3. Effect on **patient outcomes**: mortality, complications, treatment, hospital utilization

paul.goldsmith-pinkham@yale.edu · <https://paulgp.com>

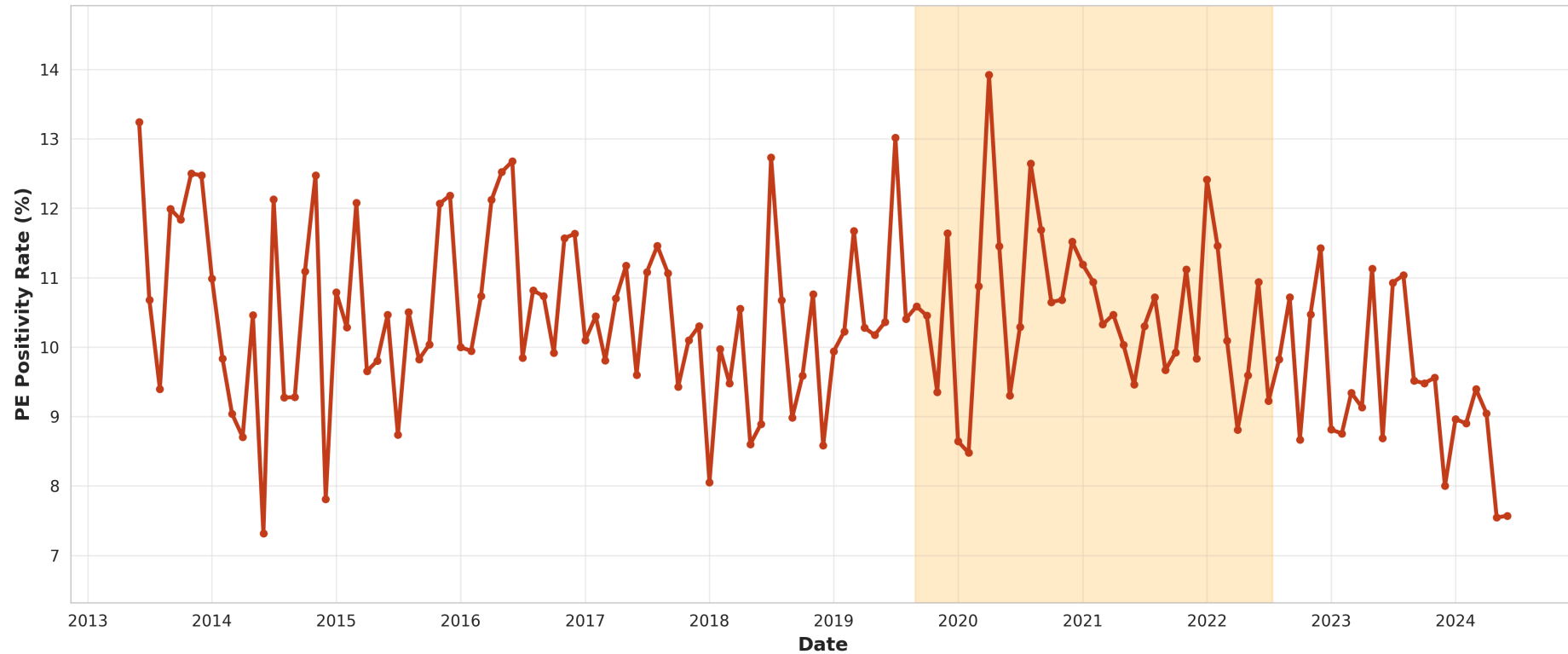


# APPENDIX

---



Same patterns hold overall in the rollout window



Stable at 7–14% across the rollout window — AI did not mechanically shift the diagnosis rate.

- 8 care sites that received AI
- Sites 1–3: 75% of volume; rolled in Aug–Sep 2019
- Sites 4–8: rolled in Jul 2020 – Jul 2022
- Volume per site: 200 to 47,175
- System-wide PE positivity: 10.2% (range: 6.9–11.3%)

## **Sample totals**

- 117,063 CTPA scans
- 84,640 patients
- 388 signing radiologists

- $N = 84\{, \}640$  patients; even split pre-AI / AI period
- 60% female; 69% White; mean age 60
- 78% receive a single scan
- PE diagnosis rate: 12% of patients
- Mortality: 5% / 7% / 12% at 30 / 90 / 365 days

Demographics stable across periods (< 3 pp differences). Largest change: 10 pp drop in patients receiving multiple scans.

## Scan-level totals

- 117,065 CTPAs
- Pre-AI: 54,150; AI: 62,915
- Volume +16% across rollout

## AI period agreement (scan-level)

- Both positive: 8.9%
- Both negative: 87.8%
- AI only: 1.7%
- Rad only: 1.6%

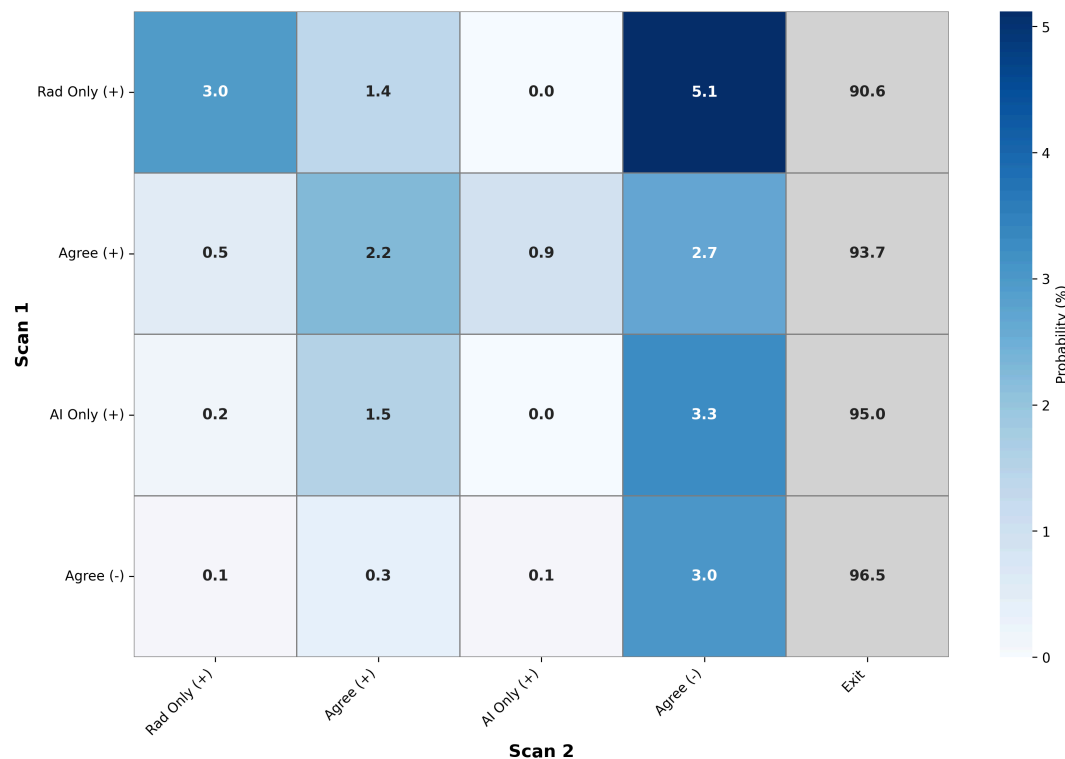
ED share rises 56% → 75%. Shift mix essentially unchanged: day 15–16% / evening 51–52% / night 33–34%.

- 148 radiologists with AI notifications
- Median: 239 notifications, IQR [31, 946]
- Hover over AI prediction in 32% of notifications (median)
- Hover duration: 1 minute (uniform across sample)
- AI alert arrives **after** rad opens scan in 99.7% of cases
  - AI as concurrent check, not advance triage
- Hover associated with **25.7 min** mean savings on AI+ cases (median 1.4 min, IQR [-3.1, 7.9])

| Group                 | $P(\text{Rad} +   \text{AI} +)$ | $P(\text{Rad} -   \text{AI} -)$ | $N$ |
|-----------------------|---------------------------------|---------------------------------|-----|
| Female                | 88.4%                           | 96.0%                           |     |
| Male                  | 82.0%                           | 97.8%                           |     |
| Diff (F – M)          | +6.4 pp**                       | -1.8 pp                         |     |
| Experience Q1         | 85.6%                           | 95.6%                           |     |
| Experience Q2 (6–10y) | 87.2%                           | 97.2%                           |     |
| Experience Q3         | 85.0%                           | 98.6%                           |     |
| Experience Q4 (21+y)  | 80.6%                           | 97.4%                           |     |
| Linear trend          | $\beta = -0.18$                 | $\beta = +0.09$                 |     |
| Volume Q1 (low)       | 83.4%                           | 94.7%                           |     |
| Volume Q4 (high)      | 85.3%                           | 98.5%                           |     |



- When AI predicts PE — override rates:
  - Day 18.5%, Evening 16.3%, Night 15.4% ( $F = 2.10, p = 0.12$ )
  - Weekend 17.0% vs Weekday 16.1% ( $t = 0.75$ )
- When AI predicts no PE — override rates: 1.7–1.8% across all categories ( $F = 0.66$ )
- Overall disagreement: 3.04% (ER) vs 4.10% (non-ER) — driven by case mix, not different rad behavior



Exit rates: 90.6% (Rad Only +) to 96.5% (Agree -). Rad overrides → more follow-up scans ordered.